

q_1c and q_2c are homotopic to the identity. The composition $(f \vee g)_*(i_{1*} \oplus i_{2*})$ sends $(x, 0)$ to $f_*(x)$ and $(0, y)$ to $g_*(y)$ since $(f \vee g)i_1 = f$ and $(f \vee g)i_2 = g$. Hence (x, y) is sent to $f_*(x) + g_*(y)$. Combining these facts, we see that the composition across the top of the diagram is $x \mapsto f_*(x) + g_*(x)$. But this composition is also $(f + g)_*$ since $f + g = (f \vee g)c$. This finishes the proof in the covariant case.

The contravariant case is similar, using the corresponding diagram with arrows reversed. The inverse of $i_1^* \oplus i_2^*$ is $q_1^* \oplus q_2^*$ by the same reasoning. An element u in the right-hand group $h(K)$ maps to the element $(f^*(u), g^*(u))$ in the lower group $h(\Sigma X) \oplus h(\Sigma X)$ since $(f \vee g)i_1 = f$ and $(f \vee g)i_2 = g$. An element $(x, 0)$ in the lower group in the diagram maps to the element x in the left-hand group since q_1c is homotopic to the identity, and similarly $(0, y)$ maps to y . Hence (x, y) maps to $x + y$ in the left-hand group. We conclude that $u \in h(K)$ maps by the composition across the top of the diagram to $f^*(u) + g^*(u)$ in $h(\Sigma X)$. But this composition is $(f + g)^*$ by definition. $\square$

Returning to the proof of the theorem, we see that the cellular chain complexes for $h_*(X)$ and $H_*(X; G)$ are isomorphic, so we obtain isomorphisms $h_n(X) \approx H_n(X; G)$ for all n . To verify that these isomorphisms are natural with respect to maps $f: X \rightarrow Y$ we may first deform such a map f to be cellular. Then f takes each pair (X^n, X^{n-1}) to the pair (Y^n, Y^{n-1}) , hence f induces a chain map of cellular chain complexes in the h_* theory, as well as for $H_*(-; G)$. To compute these chain maps we may pass to the quotient maps $X^n/X^{n-1} \rightarrow Y^n/Y^{n-1}$. These are maps of the form $\bigvee_\alpha S_\alpha^n \rightarrow \bigvee_\beta S_\beta^n$, so the induced maps f_* on h_n are determined by their component maps $f_*: S_\alpha^n \rightarrow S_\beta^n$. This is exactly the same situation as with the cellular boundary maps before, where we saw that the degree of a map $S^n \rightarrow S^n$ determines the induced map on h_n . We conclude that the cellular chain map induced by f in the h_* theory agrees exactly with the cellular chain map for $H_*(-; G)$. This implies that the isomorphism between the two theories is natural.

The situation for cohomology is quite similar, but there is one point in the argument where a few more words are needed. For cohomology theories the cellular cochain groups are the direct product, rather than the direct sum, of copies of the coefficient group $G = h^0(\text{point})$, with one copy per cell. This means that when there are infinitely many cells in a given dimension, it is not automatically true that the cellular coboundary maps are uniquely determined by how they map factors of one direct product to factors of the other direct product. To be precise, consider the cellular coboundary map $d_n: h^n(X^n, X^{n-1}) \rightarrow h^{n+1}(X^{n+1}, X^n)$. Decomposing the latter group as a product of copies of G for the $(n+1)$ -cells, we see that d_n is determined by the maps $h^n(X^n/X^{n-1}) \rightarrow h^n(S_\alpha^n)$ associated to the attaching maps φ_α of the cells e_α^{n+1} . The thing to observe is that since φ_α has compact image, meeting only finitely many n -cells, this map $h^n(X^n/X^{n-1}) \rightarrow h^n(S_\alpha^n)$ is *finitely supported* in the sense that

there is a splitting of the domain into a product of finitely many factors and a product of the remaining possibly infinite number of factors, such that the map is zero on the latter product. Finitely supported maps have the good property that they are determined by their restrictions to the G factors of $h^n(X^n/X^{n-1})$. From this we deduce, using the lemma, that the cellular coboundary maps in the h^* -theory agree with those in ordinary cohomology with G coefficients. This extra argument is also needed to prove naturality of the isomorphisms $h^n(X) \approx H^n(X; G)$.

This completes the proof of Theorem 4.59. $\square$

Proof of Theorem 4.57: The functors $h^n(X) = \langle X, K(G, n) \rangle$ define a reduced cohomology theory, and the coefficient groups $h^n(S^i) = \pi_i(K(G, n))$ are the same as $\tilde{H}^n(S^i; G)$, so Theorem 4.59, translated into reduced cohomology, gives natural isomorphisms $T: \langle X, K(G, n) \rangle \rightarrow \tilde{H}^n(X; G)$ for all CW complexes X .

It remains to see that $T([f]) = f^*(\alpha)$ for some $\alpha \in \tilde{H}^n(K(G, n); G)$, independent of f . This is purely formal: Take $\alpha = T(\mathbb{1})$ for $\mathbb{1}$ the identity map of $K(G, n)$, and then naturality gives $T([f]) = T(f^*(\mathbb{1})) = f^*T(\mathbb{1}) = f^*(\alpha)$, where the first f^* refers to induced homomorphisms for the functor h^n , which means composition with f . $\square$

The fundamental class $\alpha = T(\mathbb{1})$ can be made more explicit if we choose for $K(G, n)$ a CW complex K with $(n-1)$ -skeleton a point. Denoting $\langle X, K(G, n) \rangle$ by $h^n(X)$, then we have

$$h^n(K) \approx h^n(K^{n+1}) \approx \text{Ker } d: h^n(K^n) \rightarrow h^{n+1}(K^{n+1}, K^n)$$

The map d is the cellular coboundary in h^* cohomology since we have $h^n(K^n) = h^n(K^n, K^{n-1})$ because K^{n-1} is a point and h^* is a reduced theory. The isomorphism of $h^n(K)$ with $\text{Ker } d$ is given by restriction of maps $K \rightarrow K$ to K^n , so the element $\mathbb{1} \in h^n(K)$ defining the fundamental class $T(\mathbb{1})$ corresponds, under the isomorphism $h^n(K) \approx \text{Ker } d$, to the inclusion $K^n \hookrightarrow K$ viewed as an element of $h^n(K^n)$. As a cellular cocycle this element assigns to each n -cell of K the element of the coefficient group $G = \pi_n(K)$ given by the inclusion of the closure of this cell into K . This means that the fundamental class $\alpha \in H^n(K; G)$ is represented by the cellular cocycle assigning to each n -cell the element of $\pi_n(K)$ given by a characteristic map for the cell.

By naturality of T it follows that for a cellular map $f: X \rightarrow K$, the corresponding element of $H^n(X; G)$ is represented by the cellular cocycle sending each n -cell of X to the element of $G = \pi_n(K)$ represented by the composition of f with a characteristic map for the cell.

The natural isomorphism $H^n(X; G) \approx \langle X, K(G, n) \rangle$ leads to a basic principle which reappears many places in algebraic topology, the idea that the occurrence or nonoccurrence of a certain phenomenon is governed by what happens in a single special case, the *universal example*. To illustrate, let us prove the following special fact:

- The map $H^1(X; \mathbb{Z}) \rightarrow H^2(X; \mathbb{Z})$, $\alpha \mapsto \alpha^2$, is identically zero for all spaces X .

By taking a CW approximation to X we are reduced to the case that X is a CW complex. Then every element of $H^1(X; \mathbb{Z})$ has the form $f^*(\alpha)$ for some $f: X \rightarrow K(\mathbb{Z}, 1)$, with α a fundamental class in $H^1(K(\mathbb{Z}, 1); \mathbb{Z})$, further reducing us to verifying the result for this single α , the ‘universal example.’ And for this universal α it is evident that $\alpha^2 = 0$ since S^1 is a $K(\mathbb{Z}, 1)$ and $H^2(S^1; \mathbb{Z}) = 0$.

Does this fact generalize? It certainly does not hold if we replace the coefficient ring $\mathbb{Z}$ by $\mathbb{Z}_2$ since $H^*(\mathbb{RP}^\infty; \mathbb{Z}_2) = \mathbb{Z}_2[x]$. Indeed, the example of $\mathbb{RP}^\infty$ shows more generally that the fundamental class $\alpha \in H^n(K(\mathbb{Z}_2, n); \mathbb{Z}_2)$ generates a polynomial subalgebra $\mathbb{Z}_2[\alpha] \subset H^*(K(\mathbb{Z}_2, n); \mathbb{Z}_2)$ for each $n \geq 1$, since there is a map $f: \mathbb{RP}^\infty \rightarrow K(\mathbb{Z}_2, n)$ with $f^*(\alpha) = x^n$ and all the powers of x^n are nonzero, hence also all the powers of α . By the same reasoning, the example of $\mathbb{CP}^\infty$ shows that the fundamental class $\alpha \in H^{2n}(K(\mathbb{Z}, 2n); \mathbb{Z})$ generates a polynomial subalgebra $\mathbb{Z}[\alpha]$ in $H^*(K(\mathbb{Z}, 2n); \mathbb{Z})$. As we shall see in [SSAT], $H^*(K(\mathbb{Z}, 2n); \mathbb{Z})/\text{torsion}$ is exactly this polynomial algebra $\mathbb{Z}[\alpha]$.

A little more subtle is the question of identifying the subalgebra of $H^*(K(\mathbb{Z}, n); \mathbb{Z})$ generated by the fundamental class α for odd $n \geq 3$. By the commutativity property of cup products we know that α^2 is either zero or of order two. To see that α^2 is nonzero it suffices to find a single space X with an element $\gamma \in H^n(X; \mathbb{Z})$ such that $\gamma^2 \neq 0$. The first place to look might be $\mathbb{RP}^\infty$, but its cohomology with $\mathbb{Z}$ coefficients is concentrated in even dimensions. Instead, consider $X = \mathbb{RP}^\infty \times \mathbb{RP}^\infty$. This has $\mathbb{Z}_2$ cohomology $\mathbb{Z}_2[x, y]$ and Example 3E.5 shows that its $\mathbb{Z}$ cohomology is the $\mathbb{Z}_2[x^2, y^2]$ -submodule generated by 1 and $x^2y + xy^2$, except in dimension zero of course, where 1 generates a $\mathbb{Z}$ rather than a $\mathbb{Z}_2$. In particular we can take $z = x^{2k}(x^2y + xy^2)$ for any $k \geq 0$, and then all powers z^m are nonzero since we are inside the polynomial ring $\mathbb{Z}_2[x, y]$. It follows that the subalgebra of $H^*(K(\mathbb{Z}, n); \mathbb{Z})$ generated by α is $\mathbb{Z}[\alpha]/(2\alpha^2)$ for odd $n \geq 3$.

These examples lead one to wonder just how complicated the cohomology of $K(G, n)$ ’s is. The general construction of a $K(G, n)$ is not very helpful in answering this question. Consider the case $G = \mathbb{Z}$ for example. Here one would start with S^n and attach $(n+2)$ -cells to kill $\pi_{n+1}(S^n)$. Since $\pi_{n+1}(S^n)$ happens to be cyclic, only one $(n+2)$ -cell is needed. To continue, one would have to compute generators for π_{n+2} of the resulting space $S^n \cup e^{n+2}$, use these to attach $(n+3)$ -cells, then compute the resulting π_{n+3} , and so on for each successive dimension. When $n = 2$ this procedure happens to work out very neatly, and the resulting $K(\mathbb{Z}, 2)$ is $\mathbb{CP}^\infty$ with its usual CW structure having one cell in each even dimension, according to an exercise at the end of the section. However, for larger n it quickly becomes impractical to make this procedure explicit since homotopy groups are so hard to compute. One can get some idea of the difficulties of the next case $n = 3$ by considering the homology groups of $K(\mathbb{Z}, 3)$. Using techniques in [SSAT], the groups $H_i(K(\mathbb{Z}, 3); \mathbb{Z})$ for $0 \leq i \leq 12$ can be

computed to be

$$\mathbb{Z}, 0, 0, \mathbb{Z}, 0, \mathbb{Z}_2, 0, \mathbb{Z}_3, \mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_{10}, \mathbb{Z}_2$$

To get this sequence of homology groups would require quite a few cells, and the situation only gets worse in higher dimensions, where the homology groups are not always cyclic.

Indeed, one might guess that computing the homology groups of $K(\mathbb{Z}, n)$'s would be of the same order of difficulty as computing the homotopy groups of spheres, but by some miracle this is not the case. The calculations are indeed complicated, but they were completely done by Serre and Cartan in the 1950s, not just for $K(\mathbb{Z}, n)$'s, but for all $K(G, n)$'s with G finitely generated abelian. For example, $H^*(K(\mathbb{Z}, 3); \mathbb{Z}_2)$ is the polynomial algebra $\mathbb{Z}_2[x_3, x_5, x_9, x_{17}, x_{33}, \dots]$ with generators of dimensions $2^i + 1$, indicated by the subscripts. And in general, for G finitely generated abelian, $H^*(K(G, n); \mathbb{Z}_p)$ is a polynomial algebra on generators of specified dimensions if p is 2, while for p an odd prime one gets the tensor product of a polynomial ring on generators of specified even dimensions and an exterior algebra on generators of specified odd dimensions. With $\mathbb{Z}$ coefficients the description of the cohomology is not nearly so neat, however. We will study these questions in some detail in [SSAT].

There is a good reason for being interested in the cohomology of $K(G, n)$'s, arising from the equivalence $H^n(X; G) \approx \langle X, K(G, n) \rangle$. Taking $\mathbb{Z}$ coefficients for simplicity, an element of $H^m(K(\mathbb{Z}, n); \mathbb{Z})$ corresponds to a map $\theta: K(\mathbb{Z}, n) \rightarrow K(\mathbb{Z}, m)$. We can compose θ with any map $f: X \rightarrow K(\mathbb{Z}, n)$ to get a map $\theta f: X \rightarrow K(\mathbb{Z}, m)$. Letting f vary and keeping θ fixed, this gives a function $H^n(X; \mathbb{Z}) \rightarrow H^m(X; \mathbb{Z})$, depending only on θ . This is the idea of *cohomology operations*, which we study in more detail in §4.L.

The equivalence $H^n(X; G) \approx \langle X, K(G, n) \rangle$ also leads to a new viewpoint toward cup products. Taking G to be a ring R and setting $K_n = K(R, n)$, then if we are given maps $f: X \rightarrow K_m$ and $g: Y \rightarrow K_n$, we can define the cross product of the corresponding cohomology classes by the composition

$$X \times Y \xrightarrow{f \times g} K_m \times K_n \longrightarrow K_m \wedge K_n \xrightarrow{\mu} K_{m+n}$$

where the middle map is the quotient map and μ can be defined in the following way. The space $K_m \wedge K_n$ is $(m + n - 1)$ -connected, so by the Hurewicz theorem and the Künneth formula for reduced homology we have isomorphisms $\pi_{m+n}(K_m \wedge K_n) \approx H_{m+n}(K_m \wedge K_n) \approx H_m(K_m) \otimes H_n(K_n) \approx R \otimes R$. By Lemmas 4.7 and 4.31 there is then a map $\mu: K_m \wedge K_n \rightarrow K_{m+n}$ inducing the multiplication map $R \otimes R \rightarrow R$ on π_{m+n} . Or we could use the isomorphism $H^{m+n}(K_m \wedge K_n; R) \approx \text{Hom}(H_{m+n}(K_m \wedge K_n), R)$ and let μ be the map corresponding to the cohomology class given by the multiplication homomorphism $R \otimes R \rightarrow R$.

The case $R = \mathbb{Z}$ is particularly simple. We can take S^m as the $(m+1)$ -skeleton of K_m , and similarly for K_n , so $K_m \wedge K_n$ has $S^m \wedge S^n$ as its $(m+n+1)$ -skeleton and we can obtain μ by extending the inclusion $S^m \wedge S^n = S^{m+n} \hookrightarrow K_{m+n}$.

It is not hard to prove the basic properties of cup product using this definition, and in particular the commutativity property becomes somewhat more transparent from this viewpoint. For example, when $R = \mathbb{Z}$, commutativity just comes down to the fact that the map $S^m \wedge S^n \rightarrow S^n \wedge S^m$ switching the factors has degree $(-1)^{mn}$ when regarded as a map of S^{m+n} .

Fibrations

Recall from §4.2 that a fibration is a map $p: E \rightarrow B$ having the homotopy lifting property with respect to all spaces. In a fiber bundle all the fibers are homeomorphic by definition, but this need not be true for fibrations. An example is the linear projection of a 2-simplex onto one of its edges, which is a fibration according to an exercise at the end of the section. The following result gives some evidence that fibrations should be thought of as a homotopy-theoretic analog of fiber bundles:

Proposition 4.61. *For a fibration $p: E \rightarrow B$, the fibers $F_b = p^{-1}(b)$ over each path component of B are all homotopy equivalent.*

Proof: A path $\gamma: I \rightarrow B$ gives rise to a homotopy $g_t: F_{\gamma(0)} \rightarrow B$ with $g_t(F_{\gamma(0)}) = \gamma(t)$. The inclusion $F_{\gamma(0)} \hookrightarrow E$ provides a lift $\tilde{g}_0$, so by the homotopy lifting property we have a homotopy $\tilde{g}_t: F_{\gamma(0)} \rightarrow E$ with $\tilde{g}_t(F_{\gamma(0)}) \subset F_{\gamma(t)}$ for all t . In particular, $\tilde{g}_1$ gives a map $L_\gamma: F_{\gamma(0)} \rightarrow F_{\gamma(1)}$. The association $\gamma \mapsto L_\gamma$ has the following basic properties:

- (a) If $\gamma \simeq \gamma' \text{ rel } \partial I$, then $L_\gamma \simeq L_{\gamma'}$. In particular the homotopy class of L_γ is independent of the choice of the lifting $\tilde{g}_t$ of g_t .
- (b) For a composition of paths $\gamma\gamma'$, $L_{\gamma\gamma'}$ is homotopic to the composition $L_{\gamma'}L_\gamma$.

From these statements it follows that L_γ is a homotopy equivalence with homotopy inverse $L_{\bar{\gamma}}$, where $\bar{\gamma}$ is the inverse path of γ .

Before proving (a), note that a fibration has the homotopy lifting property for pairs $(X \times I, X \times \partial I)$ since the pairs $(I \times I, I \times \{0\} \cup \partial I \times I)$ and $(I \times I, I \times \{0\})$ are homeomorphic, hence the same is true after taking products with X .

To prove (a), let $\gamma(s, t)$ be a homotopy from $\gamma(t)$ to $\gamma'(t)$, $(s, t) \in I \times I$. This determines a family $g_{st}: F_{\gamma(0)} \rightarrow B$ with $g_{st}(F_{\gamma(0)}) = \gamma(s, t)$. Let $\tilde{g}_{0,t}$ and $\tilde{g}_{1,t}$ be lifts defining L_γ and $L_{\gamma'}$, and let $\tilde{g}_{s,0}$ be the inclusion $F_{\gamma(0)} \hookrightarrow E$ for all s . Using the homotopy lifting property for the pair $(F_{\gamma(0)} \times I, F_{\gamma(0)} \times \partial I)$, we can extend these lifts to lifts $\tilde{g}_{st}$ for $(s, t) \in I \times I$. Restricting to $t = 1$ then gives a homotopy $L_\gamma \simeq L_{\gamma'}$.

Property (b) holds since for lifts $\tilde{g}_t$ and $\tilde{g}'_t$ defining L_γ and $L_{\gamma'}$ we obtain a lift defining $L_{\gamma\gamma'}$ by taking $\tilde{g}_{2t}$ for $0 \leq t \leq 1/2$ and $\tilde{g}'_{2t-1}L_{\gamma'}$ for $1/2 \leq t \leq 1$. $\square$

One may ask whether fibrations satisfy a homotopy analog of the local triviality property of fiber bundles. Observe first that for a fibration $p: E \rightarrow B$, the restriction